

MIRAMAR MCAS, CA, USA

WMO: 722931

Lat: 32.867N

Lon: 117.133W

Elev: 477

StdP: 14.44

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93107

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	39.3	41.7	10.1	9.4	62.3	15.3	12.1	62.8	19.0	59.2	16.8	59.4	5.6	80	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.8	92.1	65.8	88.5	65.6	85.2	65.5	71.7	84.2	70.4	82.3	69.2	80.4	8.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	103.1	76.7	66.1	98.0	75.5	65.0	94.1	74.2	35.6	84.7	34.5	82.5	33.5	80.6	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.6	12.8	11.5	DB	35.0	99.8	2.6	4.9	33.2	103.3	31.6	106.1	30.2	108.9	28.3	112.4
			WB	30.2	74.4	2.5	1.8	28.3	75.7	26.8	76.8	25.4	77.8	23.6	79.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	57.1	57.1	58.5	60.5	63.4	67.2	71.7	73.1	71.7	66.9	61.4	56.3	(6)
(7)		DBStd	7.70	5.31	5.45	4.92	5.18	4.15	3.90	3.99	4.03	5.16	5.56	5.70	4.71	(7)
(8)		HDD50	15	5	3	1	1	0	0	0	0	0	0	1	4	(8)
(9)		HDD65	1393	251	230	210	152	79	17	1	0	3	41	138	271	(9)
(10)		CDD50	5045	224	201	264	316	416	517	672	717	652	523	342	201	(10)
(11)		CDD65	949	5	8	8	18	30	84	208	252	205	99	29	3	(11)
(12)		CDH74	7135	133	146	153	296	262	468	1158	1654	1471	922	404	68	(12)
(13)		CDH80	2352	19	34	38	96	93	127	310	541	562	384	138	10	(13)
(14)	Wind	WSAvg	5.4	5.6	5.8	5.5	5.9	5.5	5.1	4.9	4.9	5.1	5.3	5.3	5.5	(14)
(15)	Precipitation	PrecAvg	10.6	2.0	2.0	2.0	0.9	0.2	0.1	0.0	0.0	0.2	0.4	1.3	1.4	(15)
(16)		PrecMax	28.1	8.9	11.8	7.7	3.9	1.3	0.7	0.5	1.4	1.3	2.6	5.7	4.7	(16)
(17)		PrecMin	2.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	5.9	2.1	2.4	2.0	1.0	0.3	0.2	0.1	0.2	0.3	0.6	1.4	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.4	84.7	84.4	89.6	89.7	90.1	92.1	94.3	97.1	96.3	90.5	81.2	(19)
(20)			MCWB	55.9	57.2	57.9	58.4	59.6	67.3	70.6	69.9	68.9	63.1	58.9	54.0	(20)
(21)		2%	DB	77.7	78.8	77.9	81.7	80.9	83.3	86.5	89.6	90.7	89.6	83.5	74.7	(21)
(22)			MCWB	54.3	55.4	57.2	58.2	62.6	66.9	69.2	69.4	68.7	61.4	56.7	53.6	(22)
(23)		5%	DB	73.4	73.1	73.1	76.2	75.0	79.0	83.2	85.6	86.3	83.6	78.3	71.0	(23)
(24)			MCWB	53.6	54.3	56.5	57.7	62.4	65.7	68.5	68.7	67.8	60.4	55.6	52.7	(24)
(25)		10%	DB	68.6	68.1	68.8	71.6	71.8	75.5	80.4	82.3	82.1	78.4	73.0	66.8	(25)
(26)			MCWB	52.9	54.6	56.3	58.0	60.9	64.0	67.8	68.5	67.4	61.6	56.2	53.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.6	61.8	63.6	63.8	68.1	69.4	73.1	73.4	73.5	69.5	63.5	61.4	(27)
(28)			MCDB	70.8	71.8	72.5	76.9	78.6	84.0	85.6	88.7	88.7	80.2	72.2	68.0	(28)
(29)		2%	WB	59.3	59.5	60.7	61.7	64.8	67.6	71.0	71.4	71.5	67.0	62.0	59.4	(29)
(30)			MCDB	67.5	69.9	71.0	74.4	75.6	80.5	82.5	84.4	83.9	78.9	71.6	65.6	(30)
(31)		5%	WB	57.4	58.0	59.1	60.0	62.9	66.1	69.6	70.1	70.0	65.2	60.5	57.9	(31)
(32)			MCDB	64.4	67.1	68.2	71.2	72.5	77.7	80.5	82.3	81.5	75.2	70.0	64.0	(32)
(33)		10%	WB	55.9	56.3	57.5	58.7	61.5	64.7	68.4	68.9	68.6	63.8	59.2	56.3	(33)
(34)			MCDB	63.6	64.9	65.4	68.5	69.9	74.7	78.5	80.2	78.9	73.1	68.4	62.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.5	20.3	19.3	19.3	16.1	16.4	16.7	17.8	18.7	20.4	22.2	21.1	(35)
(36)			MCDBR	29.4	29.8	27.7	30.0	23.6	22.9	21.2	22.7	25.6	31.2	30.6	28.3	(36)
(37)			MCWBR	14.0	14.0	13.6	14.0	11.8	10.2	8.5	9.2	10.3	14.3	14.4	14.7	(37)
(38)		5% WB	MCDBR	20.2	21.6	20.9	22.6	19.9	20.8	19.2	20.0	20.9	21.0	22.4	18.4	(38)
(39)			MCWBR	11.4	12.1	11.8	11.8	10.4	9.8	8.1	8.6	9.3	10.5	12.3	11.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.320	0.333	0.344	0.363	0.362	0.391	0.391	0.358	0.364	0.318	0.321	(40)	
(41)		taud	2.558	2.543	2.490	2.465	2.433	2.438	2.437	2.416	2.490	2.443	2.564	2.555	(41)	
(42)		Ebn at Noon	290	298	300	299	293	291	282	281	288	277	285	279	(42)	
(43)		Edh at Noon	26	29	33	35	37	36	36	36	32	31	25	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1048	1324	1742	2102	2221	2324	2264	2174	1879	1464	1141	949	(44)	
(45)		RadStd	65	114	95	110	158	140	142	89	96	103	72	67	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.69	N/A	N/A	+2.54	N/A	N/A	N/A	-355	N/A
(47) Regional (41 neighbors)		+1.19	N/A	-2.90	+1.35	N/A	N/A	N/A	-193	+423

Nomenclature: See separate page